CONTRACTS ERP · QUALITY REVIEW

CRM Module
Test Plan & Gap Analysis

A full acceptance pass over Leads and Sales Opportunities, plus the features the module still needs to be complete.

DATE	SCOPE	BUILD	SUITES
23 August 2026	**Application/CRM · features/CRM**	**master @ ffa069af +3 fixes**	**10 · 96 checks**

Contents

1 How to use this document

Every suite is a table of ordered steps with an explicit expected result and a tick box. Work a suite top to bottom — later steps depend on records created by earlier ones. Where a step needs a database check, the SQL is given inline; run it against the dev database with:

```
MYSQL_PWD='<dev password>' mysql -h localhost -u root -D erp_contracts -e "<query>"
```

Front end runs at `http://localhost:3000`, API at `:5001`. Sections 14 to 16 are not test steps: section 14 lists defects already known, so time is not spent re-reporting them, and sections 15 and 16 are the gap analysis.

> **What this module is.** Two entities. A **Lead** is a person — a `Party` of type `PERSON` carrying the `LEAD` role. A **Sales Opportunity** is a deal, joined to leads many-to-many through `SalesOpportunityRole`. The module has been specialised for real-estate unit sales: an opportunity carries a Project (`WorkEffortId`) and a Unit (`ProductId`), and closing a deal reserves the apartment.

2 Pre-flight setup

2.1 Accounts and permissions

You need three logins to test properly. All three profiles already exist in the development database — the accounts below are confirmed present, email-confirmed and not locked out. Obtain the passwords from whoever provisioned the accounts;

they cannot be recovered from the database.

Account	Email	Roles	Used for
Emad	eradwan1967@gmail.com	44 roles, incl. `CRM_View`	Suites B–J. *Ahmad* (41 roles) is an equivalent alternative
Gamal	mhussein@gmail.com	Exactly 9 roles, all CRM, nothing else	Confirms the module works with no unrelated roles. Ten further accounts share this exact profile: Mark, Reham, Nagy, Medhat, Eman, Yasmin, Ayman, Ahlam, Peter, Alaa
Ashraf	ashrafa@gmail.com	30 roles, **no** `CRM_View`	Suite A negative tests. Heavily privileged but still locked out of CRM, which proves the gate rather than merely proving a weak account is weak. Lighter alternatives: Naira, Enji, Reda, Maher

> **What Suite A can and cannot prove.** The eleven CRM-only accounts carry `CRM_Leads_View/Create/Edit/Delete` and `CRM_Contacts_*` in addition to `CRM_View` — but **no code reads those granular roles.** Only `CRM_View` gates anything, at the route level. Gamal therefore sees an identical surface to any `CRM_View` holder, and no test can distinguish them. Note the consequence for defect K10: permission to delete leads is already assigned to eleven users, and no delete function exists for it to unlock.

2.2 Verify reference data before you start

Every dropdown in this module is database-driven. If a list is empty during testing, the cause is almost always missing reference rows, not application code. Confirm these counts first.

```
SELECT ENUM_TYPE_ID, COUNT(*) FROM ENUMERATION
 WHERE ENUM_TYPE_ID LIKE 'CRM%' GROUP BY 1;
SELECT COUNT(*) FROM SALES_OPPORTUNITY_STAGE;
SELECT COUNT(*) FROM DATA_SOURCE;
```

Reference set	Expected	Notes
Pipeline stages	6	Prospect 10% → Qualified 25% → Proposal 50% → Negotiation 75% → Closed Won 100% / Closed Lost 0%
`CRM_ACTION_TYPE`	15	Drives the Next Action dropdown and all conditional fields
`CRM_CANCELLATION_REASON`	13	Low Budget, Wrong Number, Not Interested, Rent, …
`CRM_MEETING_TYPE`	5	Initial, Follow-Up, Demo, Negotiation, Closing
`CRM_MEETING_LOC`	3	Indoor, Outdoor, Online
Data sources	13	Must include `INDIRECT` — it drives the broker rule

You also need at least one **project with available apartments**, one party with the **sales rep** role and one with the **broker** role, or Suites E and G cannot be completed.

2.3 Test data file

Prepare an Excel file for Suite D with these exact headers — the importer matches on header text, so a renamed column silently imports as blank:

```
First Name | Last Name | Title | Email | Mobile Number | Lead source | Address1 | Address2 | City |
Country
```

Populate 10 rows: 6 valid, 1 with a blank First Name, 2 sharing one email address, and 1 with `Lead source` set to a value not in the map (e.g. "Billboard").

3 Suite A — Access & permissions

	Step	Expected result	✓
A1	Log in as the non-CRM user, navigate to `/sales-opportunities` directly by URL	Access denied / redirected. The route sits behind `RequireRole CRM_View`	☐
A2	Same user, navigate to `/leads`	Access denied	☐
A3	Same user, inspect the top header bar	The CRM entry is not rendered	☐
A4	Navigate to `/crm` by URL	Route no longer exists — the dead placeholder screen was removed. Confirm no blank page or console error	☐
A5	Log in as the CRM-only user, click CRM in the header	Lands on the Sales Pipeline board	☐
A6	Use the sub-menu to move between Sales Opportunities and Leads	Both load; the active item is highlighted and bold	☐
A7	Log out, then request `/api/salesOpportunities` with no bearer token	401 Unauthorized	☐

4 Suite B — Leads: create, edit, duplicates

	Step	Expected result	✓
B1	Leads → *New Lead*. Submit with every field empty	First Name is flagged required; nothing is saved	☐
B2	Enter First Name only, submit	Lead is created. Email, mobile and address are all optional	☐
B3	Create a lead with all fields: name, email, mobile, country, source, both address lines	Saved and visible in the grid	☐
B4	Confirm the contact details actually persisted	Run the SQL below — one email row and one mobile row must exist for that party	☐
B5	Set Lead Temperature to Hot, save, reopen the record	Hot is still selected after the round trip	☐
B6	Create a second lead using the **same email** as B3	A "Lead Already Exists" dialog appears instead of a duplicate record	☐
B7	In that dialog, choose to view the existing lead	Navigates to the Leads list filtered to that one party, then opens it in edit mode automatically	☐
B8	Press browser Back, then Forward	The duplicate filter does not stick — the grid returns to the full list	☐
B9	Edit a lead: change name, email, mobile, country and temperature, save, reopen	Every change persisted	☐
B10	Edit a lead and clear its email entirely, save, reopen	Record the actual behaviour — clearing a contact mech is a common weak spot	☐
B11	Enter a malformed email such as `abc@`	Record the behaviour: the field is typed `email` client-side but the server does not validate format	☐
B12	Enter an Arabic name and a +20 mobile number	Both round-trip without mangling	☐

```
SELECT p.PARTY_ID, cm.INFO_STRING AS email, tn.CONTACT_NUMBER AS phone, ds.DATA_SOURCE_ID
FROM PARTY p
LEFT JOIN PARTY_CONTACT_MECH pcm ON pcm.PARTY_ID = p.PARTY_ID
LEFT JOIN CONTACT_MECH cm        ON cm.CONTACT_MECH_ID = pcm.CONTACT_MECH_ID
LEFT JOIN TELECOM_NUMBER tn      ON tn.CONTACT_MECH_ID = cm.CONTACT_MECH_ID
LEFT JOIN PARTY_DATA_SOURCE ds   ON ds.PARTY_ID = p.PARTY_ID
WHERE p.PARTY_ID = '<new party id>';
```

5 Suite C — Leads: grid & lookup

The grid is served by OData (`/odata/leadRecords`) and paged *server-side*. Sorting and filtering must therefore act on the whole table, not just the visible page — that is what steps C3 and C4 are testing.

	Step	Expected result	✓
C1	Open Leads with more than 10 records present	10 rows shown, total count correct in the pager	☐
C2	Page forward and back; change page size	Rows change, no duplicates across pages, count stays stable	☐
C3	Sort by Name descending while on page 2	The whole dataset re-sorts — verify the true last name alphabetically appears first, not merely the page's	☐
C4	Filter the name column by a value that only exists on a later page	The match is found — proves the filter reaches the server	☐
C5	Click a lead name in the grid	Opens that lead in edit mode with all fields populated	☐
C6	Click an email cell, then a phone cell	`mailto:` and `tel:` links fire	☐
C7	Confirm a lead with no email/phone renders	Shows a dash placeholder, not an empty or broken cell	☐
C8	Open an opportunity form and type 3 letters into the Leads picker	Server-side search returns matches; the list is not filtered client-side	☐
C9	Type a string matching nothing	"No leads found" is displayed	☐
C10	Search the picker by email fragment rather than name	Matches — the LOV searches both	☐

6 Suite D — Excel / CSV bulk import

This is the highest-risk area of the module: it is the only place that writes many parties in one transaction, and it is the only place with partial-failure semantics.

	Step	Expected result	✓
D1	Leads → *Upload Excel*. Drag the prepared file onto the drop zone	Parses and switches to the staging grid	☐
D2	Repeat using the click-to-browse path	Same result	☐
D3	Upload a `.csv` and a `.xls`	Both accepted — all three formats are configured	☐
D4	Upload a `.pdf` or `.docx`	Rejected by the drop zone	☐
D5	Upload a file whose sheet has 5 data rows followed by 200 visually blank rows	Exactly 5 rows appear — blank-row stripping works	☐
D6	Upload a file with a header row only	A clear "no data rows" error, not a silent empty grid	☐
D7	In the staging grid, edit a cell and delete a row	Both work before anything is committed	☐
D8	Check the row whose Lead source was "Billboard"	Falls back to `OTHER` rather than failing the row	☐
D9	Check a row whose Lead source was "facebook" in lower case	Maps to `FACEBOOK` — the match is case-insensitive	☐
D10	Click Save	Result dialog reports total received / successful / failed	☐
D11	Read the failure list	The blank-First-Name row is listed by row index with a readable reason	☐
D12	Check the two rows sharing an email	Confirm the documented behaviour for in-batch duplicates and record which of the two won	☐
D13	Confirm the good rows really committed	The successful count matches new `LEAD` rows in the database — see SQL below	☐
D14	Fix the failed row in the grid and press Save again	Only the fixed row is created; the already-saved rows are not duplicated	☐
D15	Import a file where *every* row is invalid	Zero rows created and the transaction rolls back cleanly	☐
D16	Import roughly 500 rows	Completes without timeout; note the elapsed time	☐
D17	Import a row whose email already belongs to an existing lead	Record the behaviour — cross-batch duplicate handling differs from the single-create path	☐
D18	Import Arabic names and Egyptian mobile numbers with a leading zero	Encoding preserved; the leading zero is not eaten by numeric coercion	☐

```
SELECT COUNT(*) FROM PARTY_ROLE WHERE ROLE_TYPE_ID = 'LEAD';
SELECT p.PARTY_ID, pe.FIRST_NAME, pe.LAST_NAME, p.CREATED_STAMP
FROM PARTY p JOIN PERSON pe ON pe.PARTY_ID = p.PARTY_ID
WHERE p.CREATED_STAMP >= CURDATE() ORDER BY p.CREATED_STAMP DESC;
```

7 Suite E — Opportunities: create & edit

	Step	Expected result	✓
E1	*New Opportunity*, submit empty	Stage, Owner and at least one linked Lead are all enforced	☐
E2	Fill everything except the Leads picker, submit	Inline error: at least one lead must be linked	☐
E3	Select a Project	The Unit dropdown populates with that project's apartments only	☐
E4	Pick a Unit, then change the Project	The Unit selection clears — it must not keep a unit from the old project	☐
E5	Pick a lead whose data source is **not** `INDIRECT`	The Broker field stays hidden	☐
E6	Pick a lead whose data source **is** `INDIRECT`	The Broker field appears and becomes required	☐
E7	Switch back to a non-indirect lead	Broker hides again and no stale broker is submitted	☐
E8	Use *Add New Lead* inside the opportunity form	Modal opens, the lead is created, and it is selected in the picker on close	☐
E9	Inside that modal, open the Country dropdown	The dropdown renders above the modal, not clipped behind it	☐
E10	Save a complete opportunity	Created; appears on the board in the chosen stage column	☐
E11	Verify the owner and broker links	Rows exist in `SALES_OPPORTUNITY_ROLE` with role types `OWNER`, `BROKER`, `LEAD_CONTACT`	☐
E12	Verify the party roles were auto-created	If the chosen party lacked the `BROKER` role it is added to `PARTY_ROLE` automatically	☐
E13	Edit the opportunity: change Project, Unit and Owner, save, reopen	All three persisted	☐
E14	Edit and change only the stage	Project, Unit, Owner and Leads are all untouched	☐
E15	Try to select two leads in the picker	Only the last one is kept — see defect K3, this is expected for now	☐

8 Suite F — Pipeline board & list

	Step	Expected result	✓
F1	Open the Sales Pipeline board	Six columns in sequence order, each with a count chip	☐
F2	Check a card's contents	Shows lead name, probability, owner, broker if present, close date if set, and a lead-count chip	☐
F3	Hover the lead-count chip	Tooltip lists every linked lead by name	☐
F4	Drag a card from Prospect to Qualified	Card moves, both column counts update, change persists after a page refresh	☐
F5	Drag a card and drop it back on its own column	No update call is made and no history row is written	☐
F6	Confirm probability tracked the move	Prospect 10 → Qualified 25 — the stage default is applied	☐
F7	Switch to List view	Same records in a Kendo grid with paging, sorting and filtering	☐
F8	Inspect the Project and Unit columns	Values are currently swapped — see defect K5, expected for now	☐
F9	Use Edit and Action from the list rows	Both open the same screens as the board	☐
F10	Toggle board ↔ list several times	No stale data, no duplicated fetches	☐
F11	Open a card whose opportunity has no linked lead rows	Record the behaviour — the list view reads `leads[0]` without a guard	☐

9 Suite G — Action engine

This is the core business logic. Test the matrix below **row by row**: for each action type, confirm the listed fields appear, that saving is blocked while a required one is empty, and that the stated side effects occur.

Action type	Date	Cancel reason	Meeting fields	Project + Unit	Side effect on the opportunity
Follow up	required	—	—	—	none
No Answer	required	—	—	—	none
Set meeting	required	—	—	—	none
Interested	required	—	—	—	none
Fresh Stage	required	—	—	—	none
Meeting	required	—	required	—	none
Site Visit	client only	—	required	—	none
Following up after Meeting	client only	—	—	—	none
Reservation	client only	—	—	required	retargets the opportunity's project + unit
Done Deal	client only	—	—	required	won + closed, stage → Closed Won, probability 100, apartment → reserved, history row

Action type	Date	Cancel reason	Meeting fields	Project + Unit	Side effect on the opportunity
Cancellation	—	required	—	—	closed, stage → Closed Lost, probability 0, history row naming the reason
Just info / Hiring / Broker / No answer all the time	—	—	—	—	none — comment only

"client only" means the front end demands the field but the server does not. Not a data-loss risk, but it means an API caller can skip it — see defect K4.

	Step	Expected result	✓
G1	Open an opportunity's Action modal	Two tabs, Actions and History; the Actions tab is active	☐
G2	Cycle the Next Action dropdown through all 15 types	Fields appear and disappear exactly as the matrix specifies	☐
G3	Choose Follow up, leave the date empty	Save is disabled	☐
G4	Open the date picker and try to choose yesterday	Past dates are blocked	☐
G5	Save a Follow up with a date and a comment	Appears at the top of the comment feed with type, timestamp, date and comment	☐
G6	Save a Meeting with type, location and note	All three render on the feed card	☐
G7	Choose Meeting and leave Meeting Type empty, submit	Server rejects it. *Note the error text mentions "Cancel Reason" — see defect K6*	☐
G8	Choose Reservation without a Unit, submit	Rejected: unit and project are required	☐
G9	Save a Reservation against a *different* project and unit from the opportunity's current pair	The opportunity is retargeted to the new project and unit	☐
G10	Save a **Cancellation** with a reason	Card moves to Closed Lost, is flagged closed, and the history note names the reason	☐
G11	Reopen the Action modal on that cancelled opportunity	Every input and the Save button are disabled	☐
G12	Save a **Done Deal** on a different opportunity	Moves to Closed Won, flagged won and closed, probability 100	☐
G13	Check the apartment used in G12	`APARTMENT_RESERVED` — see SQL below	☐
G14	Reopen the Action modal on the won opportunity	Locked, same as G11	☐
G15	Try to drag the won card to another column	Record the behaviour — the board does not currently block dragging a closed deal	☐
G16	Log 6 actions on one opportunity	Feed is newest-first and scrolls inside the modal	☐
G17	Save an action with a comment only, no other field	Accepted, card shows the comment	☐
G18	Save an action with no comment at all	Card shows "No additional details" rather than an empty box	☐
G19	Look for an "Answered" chip on any action	It never appears — the flag exists in the model but no control sets it. See defect K7	☐

```
SELECT so.SALES_OPPORTUNITY_ID, so.OPPORTUNITY_STAGE_ID, so.IS_WON, so.IS_CLOSED,
       so.ESTIMATED_PROBABILITY, p.PRODUCT_ID, p.APARTMENT_STATUS_ID
FROM SALES_OPPORTUNITY so LEFT JOIN PRODUCT p ON p.PRODUCT_ID = so.PRODUCT_ID
WHERE so.SALES_OPPORTUNITY_ID = '<id>';
```

10 Suite H — History & audit

	Step	Expected result	✓
H1	Open the History tab on a newly created opportunity	One entry: "Opportunity created"	☐
H2	Drag it through three stages, then reopen History	Three further entries, each naming the destination stage	☐
H3	Check the attribution line on each entry	Shows the modifying user and a timestamp	☐
H4	Log a Cancellation and check History	New entry whose note includes the cancellation reason text	☐
H5	Confirm the History tab only loads when opened	Network panel shows the history call fires on tab switch, not on modal open	☐
H6	Switch between Actions and History repeatedly	Neither list loses state or double-fetches	☐
H7	Verify history is a true snapshot	Each row carries the stage, amount, probability and close date *as at that moment*	☐

11 Suite I — Arabic / RTL

	Step	Expected result	✓
I1	Switch the interface to Arabic and reload the board	Stage names show Arabic descriptions: عميل محتمل, مؤهل, عرض, تفاوض, صفقة مكتسبة, صفقة خاسرة	☐
I2	Open the Action modal in Arabic	All 15 action types, 13 reasons, 5 meeting types and 3 locations render in Arabic	☐
I3	Check icon and button alignment throughout	Icons flip to the correct side; nothing overlaps	☐
I4	Check the comment feed and History cards in Arabic	Text is right-aligned and reads correctly	☐
I5	Confirm the language header reaches the API	Requests carry `Accept-Language: ar`	☐
I6	Switch back to English mid-session	Lists re-fetch in English without a full reload	☐
I7	Check date formatting on board cards	Note that card dates are hard-coded to the `ar-EG` calendar regardless of language — see defect K8	☐

12 Suite J — API & data integrity

These steps go around the UI on purpose. Several opportunity fields are persisted by the API but not yet collected by any form, so the only way to exercise them is directly.

	Step	Expected result	✓
J1	`POST /api/salesOpportunities` with name, description, estimated amount, currency, probability and close date	All values are stored on `SALES_OPPORTUNITY`, not only on the history row	☐
J2	`GET /api/salesOpportunities` and read that record back	Every field submitted in J1 is returned	☐
J3	`PUT /api/salesOpportunities/{id}` changing amount and close date only	Both change; name, description, project, unit and owner are untouched	☐

	Step	Expected result	✓
J4	`PATCH /api/salesOpportunities/{id}/stage` on that same record	Only the stage changes — **amount, name and close date must survive**. This is the key regression check for the sparse-DTO path	☐
J5	`POST` an opportunity with a stage id that does not exist	Clean failure message, no partial write	☐
J6	`POST` an action referencing a non-existent opportunity	Clean failure, nothing written	☐
J7	`POST` an action with an action type outside `CRM_ACTION_TYPE`	Rejected as invalid	☐
J8	Force a failure mid-transaction on the batch import, then check the tables	No orphan `PARTY` rows without `PERSON` or `PARTY_ROLE`	☐
J9	Run the orphan sweep below after finishing every suite	All three counts are zero	☐

```
-- leads with no person record
SELECT COUNT(*) FROM PARTY_ROLE pr LEFT JOIN PERSON pe ON pe.PARTY_ID = pr.PARTY_ID
WHERE pr.ROLE_TYPE_ID = 'LEAD' AND pe.PARTY_ID IS NULL;
-- opportunities with no linked lead
SELECT COUNT(*) FROM SALES_OPPORTUNITY so WHERE NOT EXISTS (
  SELECT 1 FROM SALES_OPPORTUNITY_ROLE r WHERE r.SALES_OPPORTUNITY_ID = so.SALES_OPPORTUNITY_ID
    AND r.ROLE_TYPE_ID NOT IN ('OWNER','BROKER'));
-- closed-stage opportunities left open
SELECT COUNT(*) FROM SALES_OPPORTUNITY
WHERE OPPORTUNITY_STAGE_ID IN ('SOSTG_CLOSED_WON','SOSTG_CLOSED_LOST') AND IS_CLOSED = 0;
```

13 Regression pack — three recent fixes

Three defects were repaired immediately before this test pass. Run these first; if any fails, stop and report before continuing with the full suites.

R1 Dead `/crm` screen removed

	Step	Expected result	✓
R1.1	Browse to `/crm`	Route is gone; no blank screen, no console error	☐
R1.2	Click CRM in the top header	Goes to the Sales Pipeline as before	☐
R1.3	Confirm the front end compiles	No unresolved-import error — the screen had been importing a file that does not exist	☐

R2 Opportunity fields now persisted

> **Background.** The create handler previously wrote only stage, project and unit to the opportunity row. Name, description, amount, currency, probability, close date, next step, data source, campaign and type were written to the *history* row only and were lost from the record itself. Both create and update now persist them.

	Step	Expected result	✓
R2.1	Create an opportunity through the UI, then read `SALES_OPPORTUNITY`	`ESTIMATED_PROBABILITY` and `CURRENCY_UOM_ID` are populated from the stage default and the fallback	☐
R2.2	Run Suite J steps J1–J3	All API-supplied fields survive on the entity	☐
R2.3	**Run J4.** Drag a card between columns on the board	The record's other fields are **not** blanked by the stage-only patch. This is the highest-risk regression of the change	☐
R2.4	Compare the first history row against the entity	They agree — history is now snapshotted from the saved entity	☐

R3 Cancelled and won deals now close

> **Background.** Only Done Deal previously had any effect, and even that set `IS_WON` without `IS_CLOSED`. A cancelled deal kept sitting in its original pipeline column indefinitely. Cancellation now closes the opportunity and moves it to Closed Lost, and the same flags are applied when a card is dragged to a closed column.

	Step	Expected result	✓
R3.1	Log a Cancellation with a reason on an open opportunity	Stage → Closed Lost, `IS_CLOSED = 1`, `IS_WON = 0`, probability 0	☐
R3.2	Check the history row it produced	Change note names the stage *and* the cancellation reason	☐
R3.3	Log a Done Deal on another opportunity	`IS_WON = 1` and `IS_CLOSED = 1` — previously only the first was set	☐
R3.4	Drag a card directly onto Closed Lost	Flags are set the same way as the action path	☐
R3.5	Drag a closed card back onto Prospect	It reopens: both flags clear and the Action modal unlocks again	☐

	Step	Expected result	✓
R3.6	After R3.5, check the apartment status	It stays `APARTMENT_RESERVED` — reopening does not release the unit. Known gap, listed at K9	☐
R3.7	Run the third query in Suite J's orphan sweep	Zero rows in a closed stage but still flagged open	☐

14 Known open defects — do not re-report

Found during code review and left unfixed by agreement. Confirm each still behaves as described rather than raising it again.

Ref	Defect	Effect	Location
K1	Command validators never execute	Every `CommandValidator` in the CRM feature is dead code — the MediatR pipeline has no validation behaviour, only logging. Handlers re-check the important rules by hand, so nothing is broken today, but the validators give false assurance	`ApplicationServiceExtensions.cs`
K2	Two divergent lead-create paths	`POST /leads` is live but unused; the UI posts to `/parties/createLead`. Two implementations to keep in step, one never exercised	`CRM/Leads/CreateLead.cs`
K3	Lead picker is single-select	Marked multiple, but selecting a second lead discards the first. The database supports many leads per opportunity; the UI cannot produce it	`LeadPicker.tsx:57`
K4	Client and server date rules disagree	The server's required-date list contains `FOLLOW_UP_AFTER_MEETING`, but the real enum value is `FOLLOWING_UP_AFTER_MEETING`, so that rule never fires. The client also demands dates for Site Visit, Reservation and Done Deal where the server does not	`CreateSalesOpportunityAction.cs`
K5	Swapped columns in list view	Project and Unit headers are attached to the opposite fields	`SalesOpportunityList.tsx:84`
K6	Wrong error text	A missing meeting type returns "Invalid Cancel Reason" — copy-paste in the validation branch	`CreateSalesOpportunityAction.cs`
K7	"Answered" flag unreachable	The entity, DTO and the feed's chip all support it, but no control ever sets it, so it is permanently false	`AddActionsModal.tsx`
K8	Hard-coded date locale	Board cards format dates as `ar-EG` whatever the chosen language	`SalesOpportunityBoard.tsx:95`
K9	Units are never released	Winning a deal reserves the apartment; reopening or cancelling never returns it to available	`Update/CreateAction handlers`
K10	No delete anywhere	Leads and opportunities can only be created and edited, though `CRM_Leads_Delete` is a seeded role	`module-wide`
K11	Unused lead attributes	`Tags`, `MarketingStatus`, `LastContactedTime` and `OrganizationPartyId` are declared on the DTO but never read or written	`LeadDto.cs`
K12	Actions do not record their own unit	A Reservation mutates the opportunity's project and unit but the action row itself stores neither, so the history of which unit was discussed when is lost	`SalesOpportunityAction.cs`

15 Missing features — within CRM

These are absences rather than defects: capabilities a sales team would reasonably expect that the module does not yet have.

15.1 Follow-up management HIGHEST VALUE

Every action can carry a next-action date, and the whole team is being asked to enter them — but nothing ever reads them back. There is no "my follow-ups due today" list, no overdue indicator, no calendar, no reminder, no notification. A sales rep has no way to find what they promised to do without opening opportunities one at a time. This is the single largest functional gap: the data is already being captured and simply has no consumer.

15.2 Pipeline analytics HIGHEST VALUE

The module has no reporting of any kind — there is not one CRM report handler in the codebase. Missing: conversion rate by stage, win/loss rate, average days in stage, sales-cycle length, lead-source effectiveness, rep leaderboard, forecast weighted by probability. The stage-history table already records every transition with a timestamp, so most of these are queries waiting to be written rather than new data capture.

15.3 Lead assignment and ownership

Opportunities have an owner, but leads do not. There is no assignment of an incoming lead to a rep, no reassignment, no round-robin or territory distribution, no unassigned-leads queue, and no first-response SLA. On a bulk import of several hundred leads there is no way to divide the work.

15.4 Lead lifecycle and conversion

Nothing ever promotes a lead. When a deal is won the person keeps only the `LEAD` role and never gains `CUSTOMER`, so the won buyer is invisible to the rest of the system as a customer. There is also no qualified/unqualified/nurture distinction — lead temperature is a two-value radio with no scoring behind it — and no archive or recycle path for dead leads.

15.5 Deal economics

The pipeline carries no money. Estimated amount and currency exist in the database and are now persisted, but no form collects them and the board's value chip is commented out. There is no link to unit list prices, no discount or negotiated-price capture, and therefore no weighted forecast. For a property business this is a significant hole — the pipeline cannot answer "what is in play this quarter".

15.6 Duplicate and data quality management

Duplicate detection is exact-match on a single field. There is no fuzzy matching on name or phone, no merge of two lead records, no bulk de-duplication, and no reconciliation between imported rows and existing parties. The Party module already contains a contact-mech match/dedup map that the CRM import does not use.

15.7 Communication logging

Every interaction has to be typed in by hand. There is no email, WhatsApp or SMS integration, no call logging or click-to-call, no templates, and no inbound capture — so a lead that replies by email leaves no trace in the system. Related: there is no consent or opt-out tracking, which matters as soon as any bulk outreach is attempted.

15.8 Smaller gaps

- **No opportunity detail page.** Everything happens in a 580-pixel modal; there is no single screen showing a deal with its leads, actions, history and documents together.
- **No attachments.** Nowhere to keep an ID copy, a signed reservation form or a floor plan against a lead or a deal.
- **No bulk operations** in either list — no bulk reassign, bulk stage move or bulk export.

- **Weak search.** Opportunity search covers name and description only, so a deal cannot be found by the lead's phone number or email — the way an inbound call actually arrives.
- **No competitor tracking**, though the `SalesOpportunityCompetitor` table exists and is unused.
- **No field-level audit.** History records stage transitions only; a changed owner, unit or amount leaves no trace.
- **No territory or branch segmentation**, so every rep sees every lead.

16 Missing features — integration with other modules

The CRM is currently a closed loop: leads go in, a stage changes, and the trail stops. Every item below is an existing ERP capability that the CRM does not yet reach.

16.1 Reserve Requests — the deal hand-off HIGHEST VALUE

The system already has a proper reservation workflow: `CreateReserveRequest` takes a buyer, an employee, a reserve amount, a payment method and cheque details, creates a `ReserveRequest` record and reserves the apartment. The CRM ignores all of it — a Done Deal simply writes `APARTMENT_RESERVED` straight onto the product row.

The consequence is that a deal won in the CRM produces a reserved unit with *no* reservation record: no buyer of record, no deposit, no payment method, nothing for finance to collect against. The two halves of the same business event are recorded in two places that do not know about each other. Winning a deal should raise a reserve request pre-filled from the opportunity.

16.2 Orders, quotes and invoices HIGHEST VALUE

The plumbing for this already exists and is entirely unused. `OrderItem` and `InvoiceItem` both carry a `SalesOpportunityId` column, and a `SalesOpportunityQuote` table exists — but **no handler anywhere writes any of them**. Nothing generates a quote from an opportunity, nothing creates a sales order from a won deal, and no invoice can be traced back to the deal that produced it. Until one of these is populated there is no revenue attribution: it is impossible to answer "which lead source actually produced money", only "which produced deals".

16.3 Sales commission HIGH

A full commission feature exists, with project commission rates and split handling. The CRM captures exactly the two parties commission depends on — the owning sales rep and, for indirect deals, the broker — and passes neither to it. Commission is calculated downstream from other data, so the attribution the CRM holds is discarded. The broker role in particular is captured with no broker commission consequence at all.

16.4 Catalog and unit availability HIGH

Unit reservation is fire-and-forget. There is no check that a unit is still available when a deal is won, so two opportunities can both close on the same apartment; no reservation expiry; and no release when a deal is cancelled or reopened (K9). Unit availability should be validated at close time and released on loss.

16.5 Party module HIGH

A won lead never becomes a customer (15.4), so the buyer cannot be selected in orders, invoices or payments without being re-created by hand — which is exactly how duplicate parties appear. The import also bypasses the Party module's existing contact-mech matching.

16.6 Accounting and collections MEDIUM

The module overview describes the CRM as "integrated with order and collection history, so a sales rep sees the whole customer relationship in one place". That integration does not exist in the code. A rep looking at a lead sees no invoices, no payments, no outstanding balance and no cheque status, even when the same person is an existing customer elsewhere in the system.

16.7 Business intelligence MEDIUM

The Power BI semantic model covers projects, revenues, expenses and the general ledger. There is no CRM fact or dimension in it — no pipeline snapshot, no stage-transition fact, no lead-source dimension. Since the CRM has no internal reporting either (15.2), the pipeline is currently invisible to management in both places.

16.8 Marketing campaigns MEDIUM

A complete campaign model exists in the domain — `MarketingCampaign` with roles, prices, promotions and notes. The opportunity has a `MarketingCampaignId` column that nothing ever fills. Attribution therefore stops at the lead source; there is no campaign-level cost, response rate or return.

16.9 Documents, notifications and HR MEDIUM

- **Documents.** QuestPDF is already configured across the solution, but no reservation form, offer letter or deal summary is generated from an opportunity.
- **Notifications.** No alert when a lead is assigned, when a follow-up falls due, or when a deal is won. Nothing pushes; the user must go looking.
- **HR.** Sales reps are parties with a role, unconnected to employee targets or quotas, so achieved-versus-target is not answerable.

17 Suggested build order

If the gap list is to be worked through, this sequence delivers value earliest and keeps each step small enough to ship on its own.

#	Item	Effort	Why first
1	Follow-up queue — "due today / overdue" per rep	Small	The data is already captured; this is a query and a screen. Immediate daily value
2	Won deal → reserve request hand-off	Medium	Closes the worst data-integrity hole: reserved units with no reservation record
3	Lead → customer conversion on win	Small	Unblocks orders, invoices and payments for won buyers; prevents duplicate parties
4	Deal amount on the form + pipeline value on the board	Small	Persistence already landed; this is form fields and re-enabling a chip
5	Pipeline dashboard — conversion, win rate, source performance	Medium	Stage history already holds everything needed
6	Lead assignment and ownership	Medium	Makes bulk import usable at scale
7	Unit availability validation and release	Small	Prevents double-selling the same apartment
8	Commission attribution from opportunity owner and broker	Medium	Connects the CRM to money already being calculated elsewhere
9	Opportunity detail page replacing the modal	Medium	Prerequisite for attachments, field audit and a richer timeline
10	Communication logging	Large	Highest ongoing value, but needs external service decisions first

Prepared from a full read of `Application/CRM`, `API/Controllers/CRM`, `client-app/src/features/CRM`, the CRM domain entities, seed data and the live reference tables in the development database. Defect references K1–K12 and regression items R1–R3 correspond to the code review of the same date.